BA2: Digital Korea

Week 9: Sentiment Analysis

Steven Denney

Korean Studies
Leiden University

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Today's Agenda

1. Week 8 review & check-in
2. What is sentiment analysis?
3. Dictionary-based approaches
4. Sentiment arithmetic: word + word + word = ?
5. Korean sentiment dictionaries
6. Application: Moon Jae-in's tweets
7. Demo: Sentiment analysis in Orange Data Mining
8. Looking ahead

Review & Check-In

Week 8 Review

Last week we covered:

- From sparse vectors (BoW) to dense vectors (embeddings)
- Word embedding concepts and nearest neighbors
- Semantic search and word similarity
- Document embeddings: representing entire documents as vectors

Key takeaway from Week 8

Embeddings capture **meaning**: words that appear in similar contexts end up near each other in vector space.

경제 − 성장 + 교육 ≈ 발전 (vector arithmetic)

Week 8 Review: Word Embeddings

The big idea: words as vectors capture semantic relationships.

한국 − 서울 + 도쿄 ≈ **일본**

Today we use a similar intuition, but instead of **meaning**, we measure **feeling**.

From meaning to sentiment

Word embeddings ask: *what does this word mean?*
Sentiment analysis asks: *how does this word feel?*

What Is Sentiment Analysis?

What Is Sentiment Analysis?

Sentiment analysis = determining whether a text expresses a positive, negative, or neutral attitude.

Examples in everyday life

- Product reviews (5 stars?)
- Social media posts (happy? angry?)
- News coverage (favorable? critical?)
- Political speeches (hopeful? alarming?)

Why it matters for research

- Track public opinion over time
- Compare tone across leaders
- Detect emotional framing
- Measure media bias

In computational text analysis

Sentiment analysis is a form of **measurement**: we assign a numeric value (valence) to text that captures its emotional direction.

Two Main Approaches

Dictionary-based (this week)

- Humans create word lists
- Each word gets a sentiment score
- Computer looks up and sums scores
- Transparent and interpretable
- No training data needed

Also called: lexicon-based, rule-based

Machine learning-based

- Train a model on labeled examples
- Model learns patterns from data
- Can capture context and nuance
- Requires training data
- Less transparent ("black box")

Also called: supervised classification

This week's focus

Dictionary-based methods — where the researcher defines the rules and the computer applies them at scale.

How Dictionary-Based Sentiment Works

1. Input text (tweet, speech, review)

$\downarrow$

2. Tokenize into individual words

$\downarrow$

3. Look up each word in the sentiment dictionary

$\downarrow$

4. Assign sentiment score to each word $(+1, -1, 0)$

$\downarrow$

5. Aggregate word scores $\rightarrow$ document score

The logic is simple: count positive words, count negative words, subtract.

A Simple Example

Input: 경제가 성장하고 국민이 행복합니다
(The economy is growing and the people are happy)

Word	In dictionary?	Score
경제 (economy)	neutral	0
성장 (growth)	**positive**	$+1$
국민 (people)	neutral	0
행복 (happiness)	**positive**	$+1$
Total		$+2$

Result

Two positive words, zero negative words → **positive sentiment** $(+2)$

Sentiment Arithmetic

Sentiment Arithmetic: Word + Word + Word = ?

Last week: **word + word = meaning** (embedding analogies)
This week: **word + word + word = feeling** (sentiment scoring)

희망	+	경제	+	발전	=	Positive
$+1$		0		$+1$		$+2$
위기	+	경제	+	불안	=	Negative
-1		0		-1		-2

Each word contributes a score. The sum determines the overall sentiment.

Your Turn: Guess the Sentiment

What is the sentiment of each combination?

1. 평화 (peace) + 협력 (cooperation) + 대화 (dialogue) = **?**

2. 전쟁 (war) + 위협 (threat) + 공포 (fear) = **?**

3. 성장 (growth) + 위기 (crisis) + 극복 (overcome) = **?**

4. 개혁 (reform) + 실패 (failure) + 희망 (hope) = **?**

Think about it

Cases 3 and 4 are **mixed**. This is where dictionary methods get interesting — and where their limitations start to show.

Sentiment Arithmetic: Answers

#	Words	Scores	Result
1	평화 + 협력 + 대화	$(+1) + (+1) + (+1)$	**Positive ($+3$)**
2	전쟁 + 위협 + 공포	$(-1) + (-1) + (-1)$	**Negative (-3)**
3	성장 + 위기 + 극복	$(+1) + (-1) + (+1)$	**Positive ($+1$)**
4	개혁 + 실패 + 희망	$(+1) + (-1) + (+1)$	**Positive ($+1$)**

Key insight

Dictionary methods treat every word independently. They don't understand *context*: "overcome a crisis" and "cause a crisis" would score the same for 위기.

When Arithmetic Breaks Down

Dictionary-based sentiment has known limitations

1. **Negation:** 행복하지 않다 (not happy) — "happy" is positive, but the sentence is negative
2. **Sarcasm:** 정말 잘하셨네요 (you did really well) — could be sincere or sarcastic
3. **Domain specificity:** 급등 (surge) — positive for stocks, negative for infections
4. **Mixed sentiment:** 어렵지만 희망적이다 (difficult but hopeful) — both negative and positive

So why use dictionaries?

Despite these limits, dictionary methods are **transparent**, **reproducible**, require **no training data**, and work well at scale when you understand their assumptions.

Korean Sentiment Dictionaries

Korean Sentiment Dictionaries

Several sentiment lexicons exist for Korean:

Dictionary	**Size**	**Notes**
KNU Korean Sentiment	$\sim$14,800 entries	Widely used in research
KOSAC	$\sim$17,500 entries	Corpus-based, multi-domain
Course dictionaries	positive + negative	Curated for this course
Custom dictionaries	variable	You can build your own!

For this course

We provide **positive.txt** and **negative.txt** in the course data folder. These are Korean-specific word lists that include emoticons, adjectives, and multi-word expressions.

Inside Our Sentiment Dictionaries

positive.txt (sample)

- 감동 (moved, touched)
- 기쁘다 (glad, happy)
- 발전 (development)
- 성공 (success)
- 희망 (hope)
- (^_^) (emoticons too!)

negative.txt (sample)

- 가난 (poverty)
- 걱정 (worry)
- 고통 (suffering)
- 실패 (failure)
- 위기 (crisis)
- 전쟁 (war)

How scoring works

Word found in positive.txt $\rightarrow +1$. Word found in negative.txt $\rightarrow -1$. Not found $\rightarrow 0$.
Document score = sum of all word scores.

Building Custom Dictionaries

Sometimes general dictionaries miss domain-specific sentiment

1. **Start with a general dictionary** (our positive/negative lists)
2. **Read a sample of your texts —** note domain-specific words
3. **Add domain terms:**
   - Political: 민주화 (democratization) → positive? 독재 (dictatorship) → negative?
   - Economic: 호황 (boom) → positive. 불황 (recession) → negative
4. **Validate:** check a sample of scored documents — do the scores make sense?

Remember GRS Chapter 2

Dictionary construction requires **domain knowledge**. You are making analytical choices about what counts as positive or negative in your research context.

Application: Moon Jae-in’s Tweets

Case Study: Moon Jae-in on Twitter

The corpus: @moonriver365

- 3,148 tweets (2012–2020), 9 columns
- Metadata: date, favorites, retweets
- Three periods: Pre-presidency (1,973), Transition (393), Presidency (782)

Moon Jae-in (문재인) — 19th President (2017–2022)

- Key topics: inter-Korean relations, COVID-19, Japan trade dispute, democracy
- Active Twitter presence, especially 2012 campaign (1,427 tweets)
- Tweeting declined sharply after taking office

Research questions:

1. Is the overall tone positive or negative?
2. Does sentiment change across the three periods?
3. Do different topics carry different sentiment?
4. How does sentiment relate to real-world events?

Sentiment Arithmetic with Real Tweets

Tweet (2018-10-17, presidency):
평화를 염원하는 우리 국민에게 큰 힘이 될 것입니다. 우리는 기필코 분단을 극복해낼 것입니다.
(It will be a great strength for our people who wish for peace. We will surely overcome division.)

Word	**Dictionary**	**Score**
평화 (peace)	positive	$+1$
염원 (wish, aspire)	positive	$+1$
힘 (strength)	positive	$+1$
분단 (division)	negative	-1
극복 (overcome)	positive	$+1$
Total		$+3$

Strongly positive — the hopeful framing outweighs the one negative word. But note: 분단 is part of what makes the tweet meaningful, not just noise.

Sentiment Arithmetic with Real Tweets

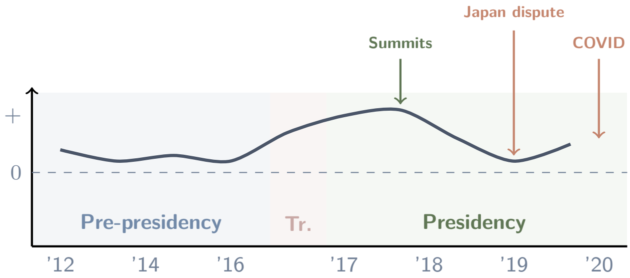
Tweet (2020-04-29, presidency):
`코로나19'를 극복하고, 이천 화재의 슬픔을 이겨내며, 반드시 우리의 새로운 일상을 만들어내겠습니다.
(We will overcome COVID-19, endure the grief of the Icheon fire, and build our new normal.)

Word	Dictionary	Score
코로나 (COVID)	—	0
극복 (overcome)	positive	$+1$
화재 (fire disaster)	negative	-1
슬픔 (grief, sadness)	negative	-1
이겨내다 (endure)	positive	$+1$
새로운 (new)	—	0
Total		0

Neutral? The dictionary says 0, but a human reads resolve and hope. This tweet responds to two tragedies with a forward-looking promise — the *rhetorical purpose* is positive, even though word-level scores cancel out.

What Can Sentiment Over Time Reveal?

Our corpus spans three distinct political periods



- Does sentiment differ across the three periods? (Your assignment!)
- Do real-world events (summits, trade dispute, COVID) produce visible shifts?
- Political communication tends toward a positive baseline — does it hold?

Moment in Time: The Impeachment Vote

Tweet (2016-12-09, transition, 4,445 favorites):

국민이 이겼습니다. 대통령 탄핵은 끝이 아니라 시작입니다.

(The people have won. The presidential impeachment is not the end but the beginning.)

Dictionary scoring:

- 이기다 (win) $\rightarrow +1$
- 탄핵 (impeachment) $\rightarrow -1$
- Score: 0 (neutral)

Context:

- Posted day after the impeachment vote
- Tone is triumphant, not neutral
- 탄핵 is negative in a dictionary but positive *in this political context*

Moment in Time: COVID Zero

Tweet (2020-04-30, presidency, 25,952 favorites — most-liked COVID tweet):

72일 만의 코로나19 국내 확진자 0명. 총선 이후 14일간 선거로 인한 감염 0명. 대한민국의 힘, 국민의 힘입니다.

(For the first time in 72 days, zero domestic COVID-19 cases. Zero election-related infections in 14 days since the general election. This is the strength of Korea, the strength of the people.)

The dictionary problem

확진자 (confirmed cases) and 감염 (infection) are negative words.
But both appear next to **0명** (zero people) — the context is celebration, not alarm.

A dictionary cannot read "zero infections" as good news. A human can.

Critical Reading: What the Dictionary Misses

Tweet (2019-08-02, presidency, 19,110 favorites):

우리는 다시는 일본에게 지지 않을 것입니다. 적지 않은 어려움이 예상되지만, 우리 기업들과 국민들에겐 그 어려움을 극복할 역량이 있습니다.

(We will never again lose to Japan. Considerable difficulties are expected, but our companies and people have the capability to overcome them.)

Dictionary says:

- 어려움 (difficulty) $\times 2 \rightarrow -2$
- 극복 (overcome) $\rightarrow +1$
- 역량 (capability) $\rightarrow +1$
- Score: 0 (neutral?)

Human reading:

- Tone is defiant and assertive
- Negative *toward Japan* specifically
- Positive *self-framing* (national strength)
- "Neutral" misses the point entirely

Lesson

Dictionary-based sentiment measures **word-level valence**. It doesn't capture the **target** (negative toward whom?) or the **rhetorical purpose** (rallying, not despairing).

Orange Data Mining Demo

Demo: Sentiment Analysis in Orange

Step 1: Load the corpus

1. Corpus widget → load `moon_twitter.csv`
2. Text column: `text` | Useful metadata: `tweet_date`, `period3`

Step 2: Preprocess

3. Python Script widget → set `TEXT_COLUMN = `text'`
4. For sentiment: include adjectives (VA) alongside nouns

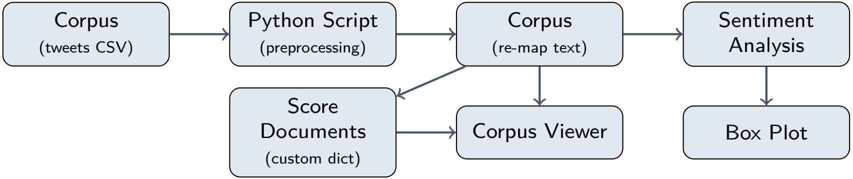
Step 3: Score sentiment

5. Sentiment Analysis widget → choose a method (e.g., Vader/Liu Hu)
6. Or: Score Documents widget → load our custom dictionaries

Step 4: Explore

7. Corpus Viewer → sort by sentiment score
8. Box Plot → compare sentiment across topics or time periods

Orange Workflow Overview



Two approaches in one workflow

- **Sentiment Analysis widget:** built-in methods (Liu Hu, Vader) — works out of the box, but designed for English
- **Score Documents widget:** load our Korean positive/negative dictionaries — better for Korean text

Using the Score Documents Widget

The **Score Documents** widget lets you score documents against custom word lists:

1. Create two word list files:
   - `positive.txt` — one positive word per line
   - `negative.txt` — one negative word per line
2. In Orange: Score Documents → select scoring method (e.g., word frequency)
3. The widget adds new columns: count of positive words, count of negative words
4. Calculate net sentiment: `positive_count` − `negative_count`

Tip

Use the **Feature Constructor** widget to create a net sentiment column: `positive - negative`

Interpreting Your Results

Once you have sentiment scores, ask:

1. **Distribution:** What is the overall shape? Mostly positive? Skewed?
2. **Outliers:** Which documents scored extremely high or low? Do they make sense when you read them?
3. **Patterns:** Does sentiment correlate with time, topic, or another variable?
4. **Validation:** Read 10–20 documents across the score range. Does the algorithm's judgment match yours?

Always validate!

Grimmer, Roberts & Stewart: "All quantitative models of language are wrong — but some are useful" (Ch. 2). Validation is not optional.

Looking Ahead

For Next Week

Week 10: Topic Modeling (LDA)

- Latent Dirichlet Allocation (LDA)
- Discovering hidden thematic structure in a corpus
- Choosing the number of topics
- Interpreting and labeling topics

Recommended reading:

- Grimmer, Roberts & Stewart — Chapter 13: Topic Models

For Next Week

Interactive exercise:

- Explore the Sentiment Analysis: Moon Jae-in's Tweets interactive on the course website
- Walk through all 6 steps — then replicate the same analysis in Orange
- R code provided in expandable sections if you want to try it in RStudio

Assignment:

- Apply sentiment analysis to the Moon Jae-in tweet corpus in Orange
- Use the Score Documents approach with our Korean dictionaries
- Compare sentiment across the three periods — does your analysis match the interactive?

Preparation:

- Orange tutorial: Text Mining #10 — Sentiment Analysis
- DataCamp: Sentiment Analysis chapter (if not yet completed)